GATE CS 2027 — Topic Weightage Map

Use this to filter any playlist, any textbook, any PYQ set. **Tier 1** = study deeply, expect questions every year. **Tier 2** = study properly, appears most years. **Tier 3** = definition/recognition level only, or skip.
Marks are indicative averages across recent GATE CS papers (total 100: 15 aptitude + 85 technical). Actual distribution varies by year and by set — treat as priority ordering, not a guarantee.

How to use this with the playlists: these playlists are titled by topic. Open the playlist, read the titles, and match them against the tier lists below. Watch all Tier 1 titles, watch Tier 2 at normal speed, skim or skip Tier 3. Do not watch in playlist order — watch in tier order within each subject.

Priority Order of Subjects

by marks per hour invested

#	Subject	Approx marks	Why this position
1	**General Aptitude**	15	Highest single block; least competitive effort from most aspirants; fastest improvement per hour
2	**Discrete Mathematics**	8–10	Largest technical block; also underpins Algorithms and TOC
3	**Algorithms**	7–9	High weight; concepts reused in DS and Discrete Maths questions
4	**Theory of Computation**	7–9	High weight, small closed syllabus, very predictable question patterns
5	**Operating Systems**	7–9	High weight; numericals are formula-driven and reliably scorable
6	**Data Structures + Programming**	8–10	Combined weight is large; C output questions are deterministic marks
7	**COA**	6–8	Cache and pipelining numericals are formulaic; rest is recall
8	**DBMS**	6–8	Narrow, well-defined question set; high return for modest effort
9	**Computer Networks**	6–8	Broad but shallow; subnetting and TCP/sliding window dominate
10	**Linear Algebra + Probability + Calculus**	5–7	Small topic lists, highly repetitive question types
11	**Digital Logic**	4–6	Small syllabus; overlaps COA
12	**Compiler Design**	4–6	Parsing is scorable; the rest has poor return per hour

Your 15-week plan should follow this order for **new** subjects, with the caveat that subjects you already covered for NMDC (OS, DBMS, CN, DS, C, COA, Digital Logic) need deepening rather than first-pass learning — so they move later in the calendar even though their weight is high.

Discrete Mathematics

~8–10 marks

Tier	Topics
Tier 1	**Propositional logic** — validity, satisfiability, tautology, logical equivalences, implication/contrapositive, CNF/DNF, inference rules · **Predicate logic** — quantifier negation, nested quantifiers, translation to/from English · **Relations** — reflexive/symmetric/transitive/antisymmetric, closures, equivalence relations, partitions, POSET, Hasse diagram, **lattice** (bounded, complemented, distributive) · **Functions** — injective/surjective/bijective, counting functions between sets · **Counting** — permutations, combinations, pigeonhole, inclusion-exclusion, balls-in-bins · **Recurrence relations** — solving linear homogeneous and non-homogeneous · **Graph theory** — degree sum, trees & spanning trees (Cayley), bipartite, planar graphs & Euler's formula, chromatic number, Euler & Hamiltonian paths, isomorphism, cut vertices/edges, matching
Tier 2	Set theory basics, power sets, cardinality · Group theory — groups, subgroups, cyclic groups, order of element, Lagrange's theorem, abelian · Generating functions · Mathematical induction · Boolean algebra as a lattice
Tier 3	Rings and fields (beyond definitions), advanced generating function manipulation, Ramsey theory, deep combinatorial identities

What actually gets asked: count the number of relations/functions/graphs with property X; determine whether a POSET is a lattice; find chromatic number; solve a recurrence; check validity of a logical argument; negate a quantified statement. Nearly every question is "count it" or "check the property" — practise those two verbs.

Algorithms

~7–9 marks

Tier	Topics
Tier 1	**Asymptotic analysis** — O/Ω/Θ/o/ω, ordering growth rates, comparing functions · **Recurrence solving** — Master theorem (all cases + inapplicable cases), substitution, recursion tree · **Complexity of code fragments** (nested loops, recursive functions) · **Sorting** — all complexities, stability, in-place, worst-case inputs, partition behaviour in Quick sort · **Greedy** — MST (Prim, Kruskal), Dijkstra, Huffman, fractional knapsack, activity selection, and **when greedy fails** · **Dynamic programming** — 0/1 knapsack, LCS, matrix chain, Bellman-Ford, Floyd-Warshall, subproblem counting · **Graph algorithms** — BFS/DFS trees, edge classification, topological sort, SCC, shortest path properties with negative edges
Tier 2	Divide and conquer (merge/quick/binary search variants), heaps & heapify cost, order statistics / median-of-medians, amortised analysis (aggregate, accounting, potential), union-find with path compression, complexity classes P/NP/NP-hard/NP-complete and standard reductions
Tier 3	Approximation algorithms, randomised algorithms beyond randomised quicksort, network flow, string matching (KMP internals), linear programming

Theory of Computation

~7–9 marks

Tier	Topics
Tier 1	**Finite automata** — DFA/NFA construction, NFA→DFA subset construction, **minimum number of states** problems, DFA minimisation (Myhill-Nerode) · **Regular expressions** & equivalence with FA · **Closure properties** of regular, CFL, DCFL, recursive, RE languages — know the full table · **Pumping lemma** for regular and CFL — applying it to prove non-membership · **CFG** — derivations, parse trees, ambiguity, CNF/GNF, simplification · **PDA** — deterministic vs non-deterministic, CFL correspondence · **Decidability & undecidability** — which problems are decidable for which class (the grid), halting problem, Rice's theorem, recursive vs recursively enumerable, closure under complement
Tier 2	Turing machine variants and equivalence, Chomsky hierarchy, reductions between problems, countability arguments, Myhill-Nerode as a proof technique
Tier 3	Linear bounded automata detail, context-sensitive grammars beyond definition, complexity-theoretic TM constructions

Highest-frequency question in the whole subject: given a language, classify it (regular / DCFL / CFL / recursive / RE / none) and decide a property. Build the closure-properties table and the decidability grid on one page and drill them — they generate a large share of the marks.

Operating Systems

~7–9 marks

Tier	Topics
Tier 1	**CPU scheduling** — FCFS, SJF, SRTF, RR, priority; Gantt charts, waiting/turnaround/response time numericals · **Process synchronisation** — critical section properties, semaphores, producer-consumer, readers-writers, dining philosophers, Peterson's algorithm, counting semaphore value tracing · **Deadlock** — four conditions, resource allocation graph, **Banker's algorithm**, safe sequences, minimum resources for deadlock-freedom · **Memory management** — paging, page table size calculations, multi-level page tables, **TLB effective access time**, segmentation, virtual memory · **Page replacement** — FIFO, LRU, Optimal, page-fault counting, Belady's anomaly
Tier 2	Threads vs processes, fork() output counting, context switching, disk scheduling algorithms and head-movement numericals, file systems and inode/block address calculations, thrashing and working set, I/O and interrupts
Tier 3	Distributed OS, real-time scheduling detail, specific OS implementation internals, security/protection models, Linux shell commands (not in the GATE syllabus — drop what you learned for NMDC)

Data Structures & C Programming

~8–10 marks combined

Tier	Topics
Tier 1	**C: pointers** — pointer arithmetic, pointer-to-pointer, arrays vs pointers, function pointers, structures with pointers · **Output prediction** — operator precedence, side effects, pre/post increment, short-circuit evaluation · **Recursion** — tracing, counting calls, recursive output problems · **Parameter passing** — value, reference, and the arrays-as-pointers consequence · **Trees** — BST operations and height analysis, traversals and reconstruction, counting trees with given properties (Catalan), AVL rotations · **Heaps** — array representation, build-heap cost, insert/delete tracing · **Hashing** — collision resolution, probe sequences, load factor, expected probes · **Stacks & queues** — postfix evaluation, implementing one with the other, sequence feasibility
Tier 2	Linked list manipulation and pointer-reversal code, graph representations and space/time trade-offs, binary tree node-count formulas, B/B+ tree order and height, dynamic memory and scope/storage classes, union vs struct sizing
Tier 3	File handling in C, preprocessor detail, bitfields, tries and suffix structures, skip lists, red-black tree internals (know properties only)

COA

~6–8 marks

Tier	Topics
Tier 1	**Cache** — direct/associative/set-associative mapping, tag-index-offset splitting, **average memory access time**, hit ratio, multi-level cache, write-through vs write-back · **Pipelining** — speedup, throughput, efficiency, stalls, hazards (RAW/WAR/WAW), operand forwarding, branch penalty calculations · **Addressing modes** and instruction formats · **Number representation** — 2's complement, range, overflow, IEEE 754 encoding and decoding
Tier 2	Memory interleaving, secondary storage and disk access time, machine instruction cycles and micro-operations, control unit (hardwired vs microprogrammed), I/O: DMA, interrupts, polling, RISC vs CISC
Tier 3	Booth's multiplication and restoring/non-restoring division traces, floating-point arithmetic algorithms, cache coherence protocols, superscalar/VLIW, specific processor architectures

DBMS

~6–8 marks

Tier	Topics
Tier 1	**Normalization** — FD closure, **finding all candidate keys**, 2NF/3NF/BCNF determination, lossless join, dependency preservation, canonical cover · **Transactions** — **conflict serializability** and precedence graphs, view serializability, recoverable/cascadeless/strict schedules, 2PL and variants, timestamp ordering · **SQL** — output of a given query, joins, nested/correlated subqueries, aggregates with NULL, group by/having · **Relational algebra** — expression evaluation, equivalence, division
Tier 2	B/B+ tree order, height and node-capacity calculations; indexing (dense/sparse/primary/secondary/clustering) and block-access counting; ER model to relational schema and minimum-table counting; integrity constraints
Tier 3	Query optimization and cost estimation, distributed databases, NoSQL, data warehousing/OLAP, PL/SQL, triggers and cursors, physical storage detail

Computer Networks — ~6–8 marks

Tier	Topics
Tier 1	**IP addressing & subnetting** — CIDR, VLSM, subnet/broadcast/host-range calculation, address aggregation · **Sliding window protocols** — stop-and-wait, GBN, SR; window sizes, **efficiency and throughput numericals**, utilisation · **TCP** — three-way handshake, sequence/ack numbers, congestion control (slow start, AIMD) window-size tracing, timers · **Error detection** — CRC computation, checksum, Hamming distance and code
Tier 2	Layer-wise protocol and device mapping, CSMA/CD and collision/minimum-frame-size calculations, ALOHA efficiency, routing algorithms (distance vector, link state) and convergence tracing, IPv4 header and fragmentation calculations, DNS/DHCP/ARP operation, application protocols and port numbers
Tier 3	Wireless/802.11 internals, network security specifics, SNMP, multimedia networking, ATM/MPLS, physical layer encoding detail

CN in GATE is far more **numerical** than the PSU version you prepared. Subnetting, sliding-window efficiency and TCP congestion-window tracing carry most of the marks — your NMDC-era recall tables are necessary but not sufficient here.

Engineering Maths: LA, Probability, Calculus — ~5–7 marks

Tier	Topics
Tier 1	**Linear Algebra** — rank, system of equations (consistency, number of solutions), **eigenvalues & eigenvectors**, determinant properties, trace, characteristic equation, Cayley-Hamilton · **Probability** — conditional probability, Bayes' theorem, independence, random variables, expectation and variance, uniform/binomial/Poisson/exponential/normal distributions · **Calculus** — limits, continuity, differentiability, maxima/minima, definite integrals
Tier 2	Matrix inverse and LU decomposition, vector spaces/basis/dimension, linear independence, mean-value theorems, Taylor series, multiple integrals, convergence of series
Tier 3	Numerical methods, differential equations, complex analysis, transform methods (not in the CS syllabus)

Despite the modest mark count, these three have the **best marks-per-hour of any technical area**, because the question types repeat almost verbatim year to year. Eigenvalues, rank/consistency, and Bayes appear with near-certainty. Do the PYQs first here — they teach the pattern faster than lectures do.

Digital Logic — ~4–6 marks

Tier	Topics
Tier 1	Boolean algebra & minimisation (K-map up to 4–5 variables, don't-cares), number systems & conversions, 2's complement arithmetic and overflow, combinational circuits (MUX-based function implementation, decoders, adders), **functional completeness**
Tier 2	Sequential circuits — flip-flops, conversions, state tables and state minimisation, counters (mod-N, ring, Johnson), shift registers, race-around
Tier 3	Logic families (TTL/CMOS electrical characteristics), memory device internals, ADC/DAC, timing/propagation-delay circuit analysis

Compiler Design — ~4–6 marks

Tier	Topics
Tier 1	**Parsing** — FIRST and FOLLOW sets, LL(1) table construction and conflict detection, LR(0)/SLR(1)/CLR(1)/LALR(1) item sets and conflicts, grammar classification hierarchy · **Left recursion & left factoring** · **Lexical analysis** — tokens, lexemes, patterns
Tier 2	Syntax-directed translation (S-attributed vs L-attributed), intermediate code (three-address code, quadruples/triples), symbol table, runtime environments and activation records, basic blocks and control flow graphs, local optimisation
Tier 3	Code generation algorithms and register allocation detail, data-flow analysis theory, loop optimisation detail, peephole specifics, garbage collection

Compiler Design has the **worst return per hour** in the syllabus, but **parsing alone** reliably yields marks and is self-contained. Strategy: master FIRST/FOLLOW and LR item-set construction thoroughly, treat the rest as recognition-level, and do not attempt the whole subject deeply.

General Aptitude — 15 marks — the single largest block

Tier	Topics
Tier 1	**Quantitative** — percentage, ratio & proportion, averages, time-speed-distance, time & work, profit & loss, simple & compound interest, mixtures · **Data interpretation** — tables, bar/line/pie charts · **Logical reasoning** — number/letter series, analogies, syllogisms, statement-conclusion, Venn diagrams, puzzles, cubes & dice · **Verbal** — sentence completion, grammar (subject-verb agreement, tenses, articles, prepositions), word meanings in context, reading comprehension
Tier 2	Permutation-combination and probability basics, mensuration, geometry basics, clocks and calendars, direction sense, blood relations, verbal analogies, critical reasoning
Tier 3	Static GK, current affairs, awards, national days — **not in the GATE syllabus**. Drop everything you learned for the NMDC awareness section; it earns nothing here.

Treat aptitude as a first-class subject, not a filler. At 15 marks it outweighs every technical subject individually, and GATE aptitude is more quantitative and comprehension-heavy than the PSU version. One hour daily from week 1 to week 15, without exception.

Marking Scheme — and Why Your NMDC Habit Must Change

Question type	Marks	Negative marking
MCQ (single correct)	1	**−1/3**
MCQ (single correct)	2	**−2/3**
MSQ (multiple select)	1 or 2	**None** — but no partial credit; all correct options required
NAT (numerical answer)	1 or 2	**None**

Consequences to internalise now:
1. The "guess everything, leave nothing blank" rule from your NMDC prep is **wrong for GATE MCQs**. Blind guessing has negative expected value. Guess only after eliminating at least two options.
2. **Always attempt every NAT and MSQ** — there is no penalty, so a blank is strictly worse than a guess.
3. 65 questions in 180 minutes is ~2.75 minutes each, versus ~1.2 minutes in your NMDC paper. GATE rewards sustained multi-step work, not rapid recognition. Build stamina with full 3-hour mocks, not 40-minute sets.
4. Accuracy now matters more than coverage — the reverse of the NMDC calculus.

Practice Discipline

- **PYQs are the syllabus.** The topicwise PYQ PDFs are the highest-value item in your resource list — higher than any lecture playlist. Every subject, every topic, no exceptions.
- Do PYQs **by topic immediately after finishing that topic**, not saved for the end. You will otherwise finish the syllabus without knowing which parts you actually understood.
- Maintain one error log per subject. By week 13 the error log, not the lectures, is your revision material.
- One primary source per subject. The resource list offers lectures + revision videos + NPTEL + textbook for each; **pick one**. Resource-hopping is the main way a 15-week plan fails.
- Textbooks are for **lookup when a concept resists**, using the chapter mappings in that PDF — not for cover-to-cover reading. Exception: Rosen for Discrete Maths, where working the exercises is the point.
- Full-length 3-hour mocks only in the last 3–4 weeks. Earlier mocks mostly measure what you haven't studied yet.